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Extending LLM Context Window with Adaptive Grouped Positional Encoding: A Training-Free Method

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Extending Long-Context Processing Capabilities

Training on High-Quality Long-Context Datasets at Scale

- High performance 😊
- Limited data availability 😭
- High time and hardware costs 😭

Training-Free Positional Encoding Remapping Methods

- Low-cost scalability 😊
- High flexibility 😊
- Limited performance 😭

How to improve mapping strategies for better performance of training-free methods?



Adaptive **Gro**uped **P**ositional **E**ncoding **(AdaGroPE)**

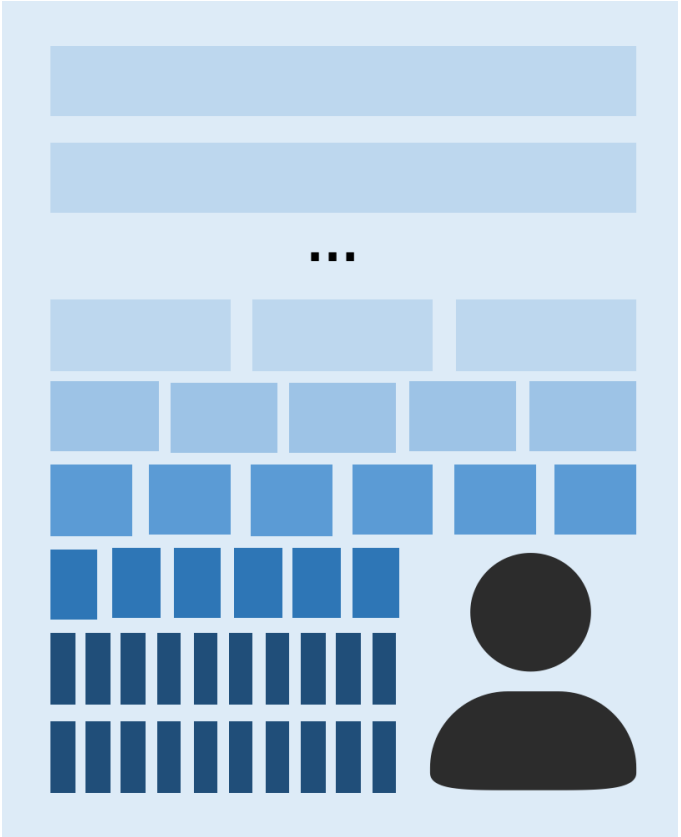
Intuitive Ideas

- Prioritize relative positions of nearby tokens, inspired by RoPE's long-distance decay design
- Align with human perception and processing patterns of long-context sequences

Plug-and-play, training-free

Easily combined with training methods

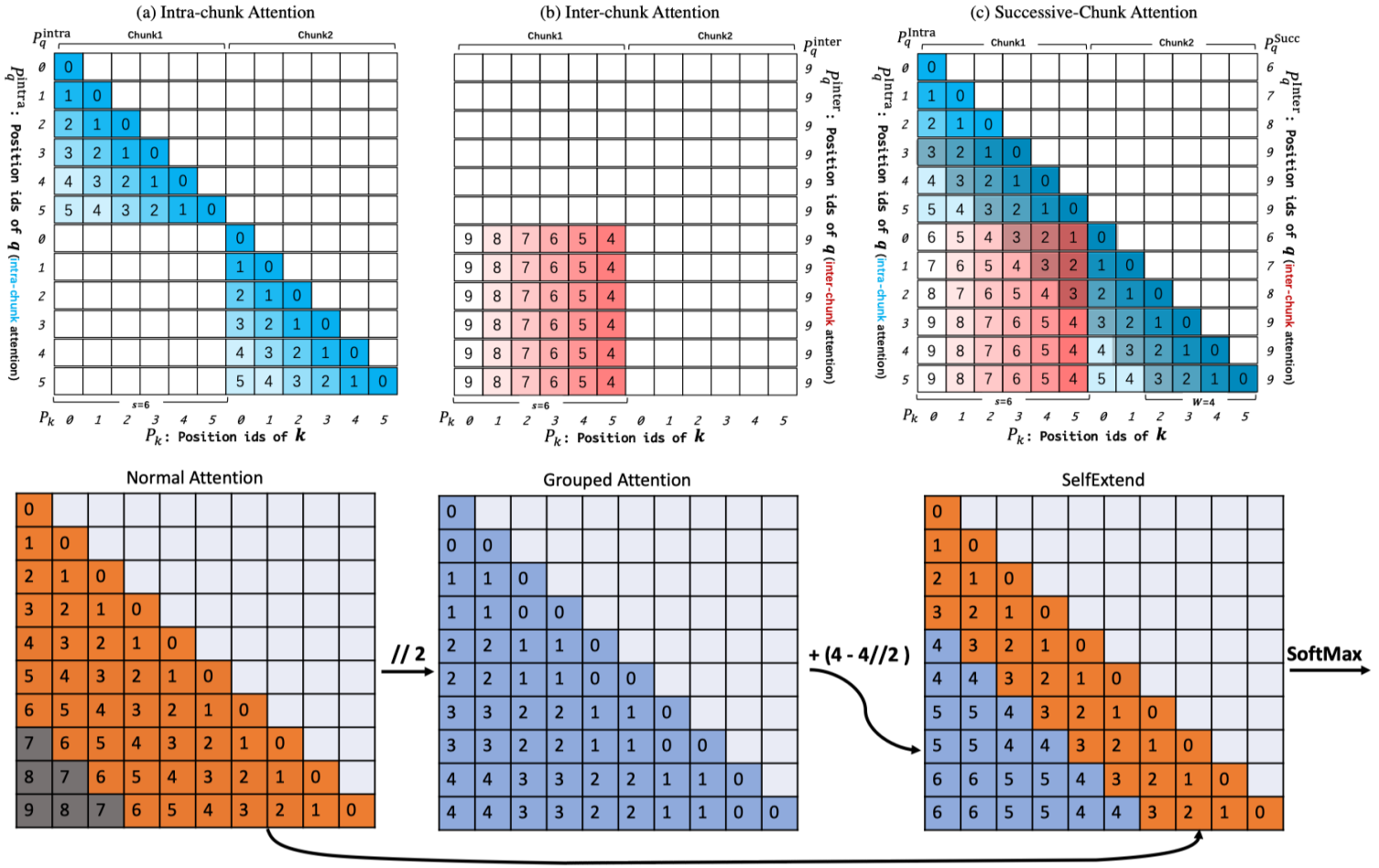
Dynamically adapts to various scenarios



Related Work

Dual Chunk Attention

SelfExtend



An C, Huang F, et al. Training-Free Long-Context Scaling of Large Language Models, 2024
Jin H, Han X, et al. LLM Maybe LongLM: SelfExtend LLM Context Window Without Tuning, 2024

AdaGroPE: An Implementation

Minimizing Reuse and Prioritizing Distant Relative Position Reuse

$$M_a[i][j] = i - j \quad \text{if } i - j < w_n \qquad M_a[i][j] = \begin{cases} i - j & \text{if } i - j < L - 2d_1, \\ f_1^m(i - j) & \text{others,} \end{cases}$$

$$\begin{aligned} L_1^{\max} &= P, \\ L_2^{\max} &= w_n + (P - w_n) \cdot 2 \\ &= 2P - w_n. \end{aligned}$$

f: mapping function from the original relative position to the AdaGroPE adjusted one

Progressive Increase in Reuse Count from Close to Distant

$$r_n = \begin{cases} \lfloor \frac{r}{n} \cdot P \rfloor & \text{if } \log_2 n \in \mathbb{N}, \\ 0 & \text{otherwise,} \end{cases}$$

$$\begin{aligned} L_3^{\max} &= r_1 + 2r_2 + (P - r_1 - r_2) \cdot 3 \\ &= 3P - 2r_1 - r_2. \end{aligned}$$

$$M_a[i][j] = \begin{cases} f_1^r(i - j) & \text{if } i - j < r_1, \\ f_2^r(i - j) & \text{if } r_1 \le i - j < L - 3d_2, \\ f_2^m(i - j) & \text{others,} \end{cases}$$

$$\begin{aligned} L_4^{\max} &= r_1 + 2r_2 + 3r_3 + (P - r_1 - r_2 - r_3) \cdot 4 \\ &= 4P - 3r_1 - 2r_2 - r_3. \end{aligned}$$

$$M_a[i][j] = \begin{cases} f_1^r(i - j) & \text{if } i - j < r_1, \\ f_2^r(i - j) & \text{if } r_1 \le i - j < r_1 + 2r_2, \\ f_3^r(i - j) & \text{if } r_1 + 2r_2 \le i - j < L - 4d_3, \\ f_3^m(i - j) & \text{others.} \end{cases}$$

Notation	Explanation
i, j	Absolute position indices
$M_a[\cdot][\cdot]$	Relative position in AdaGroPE
L	Target context length after extension
w	Pre-trained context window length
P	Number of relative positions used; $P \le w$
r	Coefficient controlling minimum retained positions per usage count
w_n	Size of neighbor window preserving original relative positions; $w_n = rP$
G_s	Usage count of relative positions; increases with position
$G_s^m(\cdot)$	Maximum usage count as a function of L with positions reused up to $G_s^m(L)$ times
$L_n^{\max}$	Maximum L for which $G_s^m(L) = n$, *i.e.*, $G_s^m(L) = n$ holds iff $L \in (L_{n-1}^{\max}, L_n^{\max}]$
d_n	Number of positions used $n{+}1$ times when $G_s^m = n{+}1$; $L - L_n^{\max}$ for $L \in (L_n^{\max}, L_{n+1}^{\max}]$
r_n	Minimum retained positions used n times when $G_s^m > n$; $r_1 = w_n$

AdaGroPE: An Implementation

Adaptive Relative Position Adjustment Strategy

$$L_n^{\max} = nP - \sum_{k=1}^{n-1}(n-k-1)r_k.$$

$$G_s^m(L) = \begin{cases} 1 & \text{if } L \le L_1^{\max}, \\ 2 & \text{if } L_1^{\max} < L \le L_2^{\max} \\ \vdots & \vdots \\ n+1 & \text{if } L_n^{\max} < L \le L_{n+1}^{\max} \end{cases}$$

$$M_a[i][j] = \begin{cases} f_1^r(i-j) & \text{if } i-j < r_1, \\ f_2^r(i-j) & \text{if } r_1 \le i-j < r_1 + 2r_2, \\ \vdots & \vdots \\ f_n^r(i-j) & \text{if } \sum_{k=1}^{n-1} kr_k \le i-j < \sum_{k=1}^{n} kr_k, \\ \vdots & \vdots \\ f_{g_l-1}^r(i-j) & \text{if } \sum_{k=1}^{g_l-2} kr_k \le i-j < \sum_{k=1}^{g_l-1} kr_k, \\ f_{g_l}^r(i-j) & \text{if } \sum_{k=1}^{g_l-1} kr_k \le i-j < L-(g_l+1)d_{g_l} \\ f_{g_l}^m(i-j) & \text{if } L-(g_l+1)d_{g_l} \le i-j, \end{cases}$$

Algorithm 1 Pseudocode for relative position computation in AdaGroPE during decoding

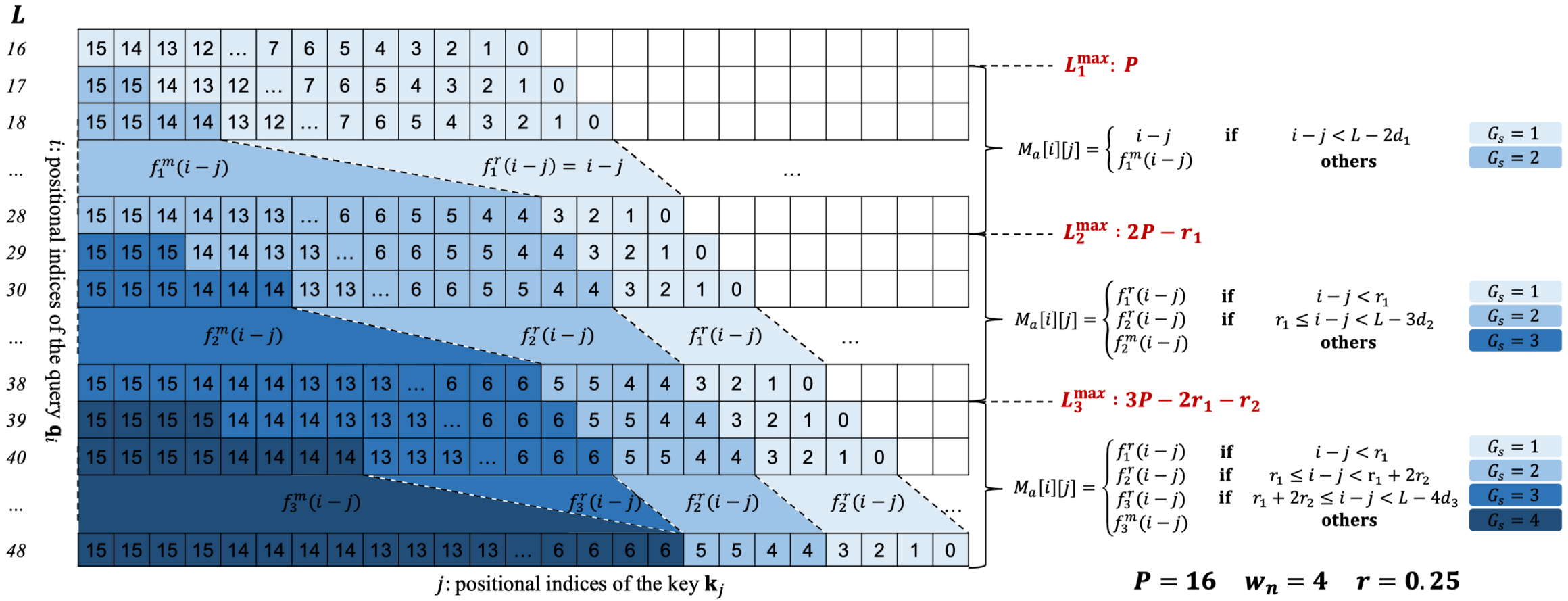
Input: L ▷ Target context length after extension
P ▷ Number of relative positions used
r ▷ Reuse ratio coefficient
Output: m_a ▷ Relative position in AdaGroPE

function AdaGroPERelPos(L, P, r)
  if $L \le P$ **then**
    $m_a \leftarrow [L-1, L-2, \ldots, 0]$
    return m_a
  end if
  $G_s^m \leftarrow 1$
  $L_n^{\max} \leftarrow P$
  $r_{\text{sum}}, l_{\text{sum}} \leftarrow 0$
  $m_a \leftarrow [\,]$
  while $L_n^{\max} < L$ **do**
    if isPowerOfTwo(G_s^m) **then**
      $r_n \leftarrow \lfloor r \times P / G_s^m \rfloor$
      for $p = r_{\text{sum}}$ to $r_{\text{sum}} + r_n - 1$ **do**
        prepend p to m_a for G_s^m times
      end for
      $r_{\text{sum}} \leftarrow r_{\text{sum}} + r_n$
      $l_{\text{sum}} \leftarrow l_{\text{sum}} + r_n \times G_s^m$
    end if
    $G_s^m \leftarrow G_s^m + 1$
    $L_n^{\max} \leftarrow (P - r_{\text{sum}}) \times G_s^m + l_{\text{sum}}$
  end while
  $d_{n-1} \leftarrow P - (L_n^{\max} - L) - r_{\text{sum}}$
  for $p = r_{\text{sum}}$ to $P - d_{n-1} - 1$ **do**
    prepend p to m_a for $G_s^m - 1$ times
  end for
  for $p = P - d_{n-1}$ to $P - 1$ **do**
    prepend p to m_a for G_s^m times
  end for
  return m_a
end function

Pseudocode for relative position computation in AdaGroPE during decoding

AdaGroPE: An Example

Relative Position Computations in AdaGroPE as L increases



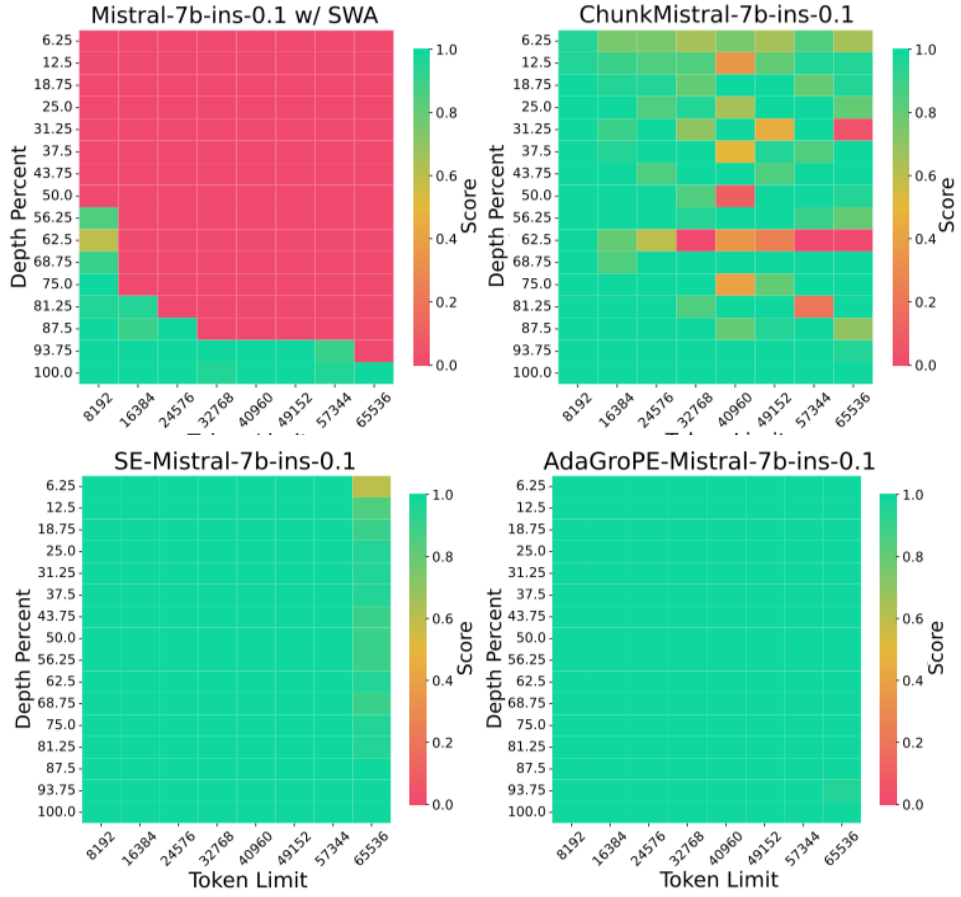
Experiments: Main Results

Language Modeling Tasks

Model	Evaluation Context Window			
	4096	8192	16384	32768
Llama-2-7b	**7.87**	>100	>100	>100
ChunkLlama-2-7b	**7.87**	7.67	7.64	7.89
SE-Llama-2-7b	**7.87**	7.67	7.58	7.71
AdaGroPE-Llama-2-7b	**7.87**	**7.65**	**7.56**	7.75
Longlora-7b-32k*	8.14	7.85	7.70	7.80
Together-7b-32k*	8.21	7.95	7.76	**7.64**
CodeLlama-7b-16k*	8.93	8.64	8.44	8.36
CLEX-7b-16k*	16.74	15.08	14.28	14.70
Llama-3-8b	**9.04**	**8.71**	78.88	>100
Chunk-Llama-3-8b	**9.04**	**8.71**	8.61	8.62
SE-Llama-3-8b	9.13	8.80	8.59	**8.52**
AdaGroPE-Llama-3-8b	**9.04**	**8.71**	**8.57**	**8.52**

AdaGroPE outperforms the baselines in most cases

Synthetic Long Context Tasks



Experiments: Main Results

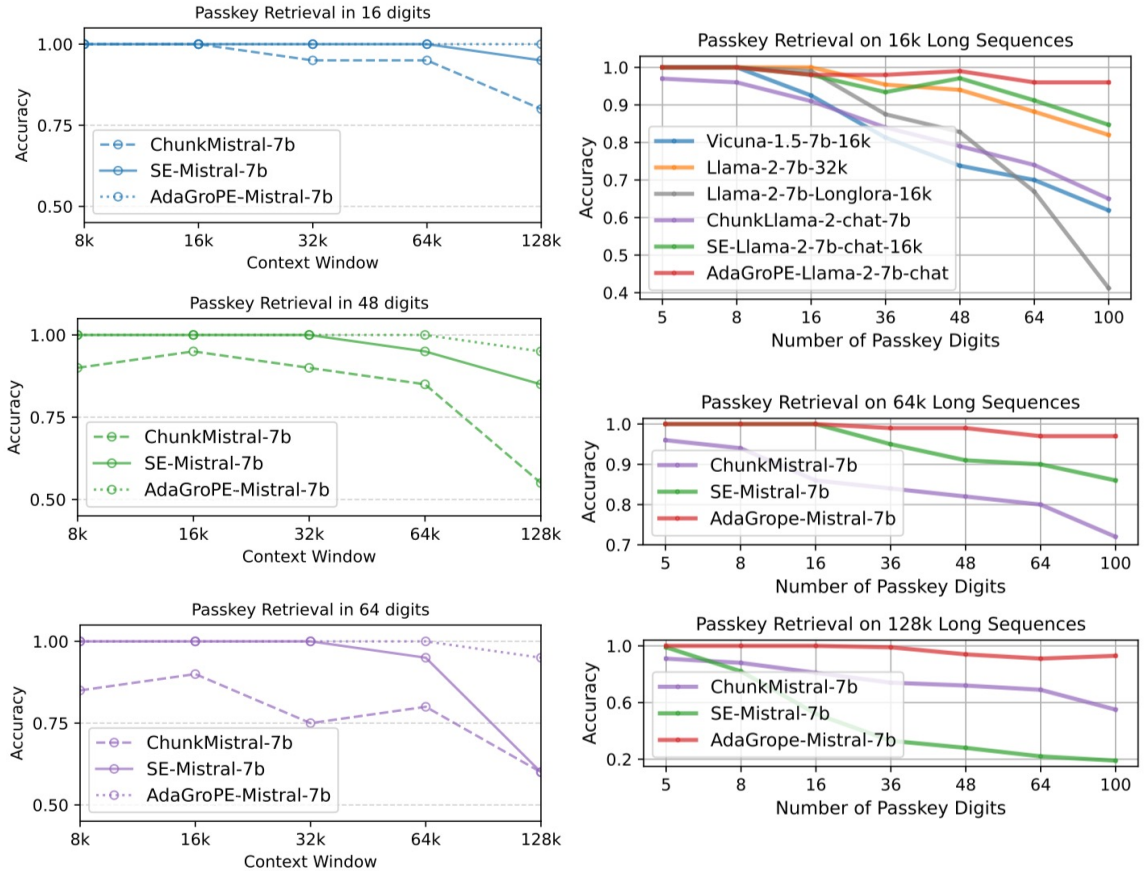
Real-World Long Context Tasks

Model	Tokens	Single-Document QA			Multi-Document QA			Summarization			Few-shot Learning			Synthetic		Code		Avg.
		NarrativeQA	Qasper	MultiField-en	HotpotQA	2WikiMQA	Musique	GovReport	QMSum	MultiNews	TREC	TriviaQA	SAMSum	PassageCount	PassageRe	Lcc	RepoBench-P	
GPT-3.5-Turbo-16k	16k	23.6	43.3	52.3	51.6	37.7	26.9	29.5	23.4	26.7	68.0	91.4	41.7	4.5	71.0	54.7	53.6	43.74
ChatGLM2-6B-32k	32k	21.1	31.5	46.2	45.1	34.0	21.9	32.4	24.0	26.5	62.5	78.7	36.3	1.5	77.0	55.6	49.9	40.26
Baichuan-13B-4k	16k	0.07	17.55	17.28	3.29	15	0.1	6.8	1.71	23.1	20.05	20.06	5.77	0.06	0.5	47.98	16.58	10.49
ALiBi-7B-4k	16k	0.04	8.13	17.87	2.73	8	1.33	5.31	1.64	25.55	9.25	8.83	4.67	0	1.27	46.69	18.54	9.48
Llama-2-7b-chat	4k	18.7	19.2	36.8	25.4	**32.8**	9.4	27.3	20.8	25.8	61.5	77.8	40.7	2.1	**9.8**	52.4	43.8	31.52
ChunkLlama-2-7b-chat	25k	20.27	25.80	34.87	23.18	28.42	10.15	27.03	21.27	**26.47**	68.50	74.86	41.51	1.48	4.75	58.05	50.76	32.34
SE-Llama-2-7b-chat	25k	**21.37**	26.68	34.63	35.47	30.46	15.51	27.51	21.30	25.87	68.50	78.79	41.29	**3.90**	3.50	**59.69**	**53.83**	34.26
AdaGroPE-Llama-2-7b-chat	25k	18.72	**29.61**	**40.20**	**37.33**	30.86	**15.73**	**28.39**	**21.45**	26.10	**69.50**	**83.23**	**42.16**	3.48	6.00	59.38	52.24	**35.27**
Llama-3-8b-ins	8k	21.63	**44.11**	44.35	46.84	35.84	21.53	29.98	22.66	27.75	75.50	90.58	42.67	6.50	66.50	56.81	51.24	42.78
ChunkLlama-3-8b-ins	25k	**26.54**	42.36	47.82	47.54	35.27	25.19	31.86	23.02	27.39	**77.00**	90.31	42.81	7.00	75.50	**58.49**	**55.15**	44.58
SE-Llama-3-8b-ins	25k	23.88	43.82	50.64	50.71	36.58	30.24	**32.90**	23.90	27.79	76.00	**91.68**	**42.93**	4.40	98.00	56.72	47.53	46.11
AdaGroPE-Llama-3-8b-ins	25k	25.72	44.09	**51.80**	**52.10**	**38.80**	**31.82**	32.81	**24.11**	**27.90**	76.50	91.13	42.30	**7.62**	**99.00**	56.49	51.46	**47.10**
Mistral-7b-ins-0.1 w/o SWA	8k	20.46	35.36	39.39	34.81	29.91	11.21	24.70	21.67	26.67	68.00	86.66	41.28	0.18	24.00	56.94	**55.85**	36.07
Mistral-7b-ins-0.1 w/ SWA	16k	19.40	34.53	37.06	42.29	32.49	14.87	27.38	22.75	26.82	65.00	**87.77**	42.34	1.41	28.50	**57.28**	53.44	37.08
ChunkMistral-7b-ins-0.1	16k	20.86	36.56	42.40	35.89	31.25	12.47	28.08	22.87	**27.09**	69.50	86.52	42.94	2.14	21.50	54.92	52.70	36.73
SE-Mistral-7b-ins-0.1	16k	23.56	**39.33**	49.50	45.28	34.92	23.14	30.71	**24.87**	26.83	69.50	86.47	**44.28**	1.18	29.50	55.32	53.44	39.86
AdaGroPE-Mistral-7b-ins-0.1	16k	**25.02**	39.00	**53.38**	**47.88**	**35.26**	**25.47**	**31.26**	23.84	26.67	**70.50**	86.66	43.86	**3.41**	**33.50**	55.05	51.50	**40.77**
SOLAR-10.7b-ins	4k	16.50	24.06	46.76	44.03	36.05	22.76	31.39	19.81	26.36	70.00	87.91	42.49	4.50	26.50	41.04	54.36	37.16
ChunkSOLAR-10.7b-ins	16k	22.48	29.77	**48.84**	51.62	34.80	27.35	31.59	21.75	26.22	**74.50**	87.41	42.69	**7.50**	20.00	48.98	54.94	39.40
SE-SOLAR-10.7b-ins	16k	22.63	32.49	47.88	46.19	34.32	27.88	30.75	22.10	25.62	**74.50**	89.04	42.79	4.00	28.00	53.73	56.47	39.90
AdaGroPE-SOLAR-10.7b-ins	16k	**24.35**	**34.42**	48.81	**53.31**	**43.30**	**33.93**	**32.38**	**22.29**	**26.51**	**74.50**	**89.62**	**43.11**	6.50	**36.00**	**54.32**	**58.64**	**42.62**
Phi-2	2k	4.46	7.01	19.98	9.43	8.55	4.62	25.64	14.32	24.03	50.50	74.55	**1.71**	**2.83**	**4.17**	**58.96**	54.14	22.81
SE-Phi-2	8k	12.04	**12.10**	20.15	8.22	9.68	3.89	27.90	14.58	22.13	**61.00**	**82.82**	1.40	2.37	2.83	57.87	**56.42**	24.71
AdaGroPE-Phi-2	8k	**14.14**	11.90	**26.80**	**9.96**	**11.37**	**5.09**	**29.68**	**20.04**	**25.19**	60.00	82.69	1.29	2.37	4.73	58.10	55.07	**26.15**

Strong performance on diverse long-context tasks and datasets

Analysis: Generalization and Compatibility

Performance as the Context and Passkey Length Increases



Integrated with Training-Based Extension Methods

Model	Position Emb	Training context	Context Window			
			32k	64k	96k	128k
CodeLlama 7b	NTK	16k	8.36	8.65	9.14	9.87
+AdaGroPE	NTK	16k	**8.34**	**8.32**	**8.38**	**8.48**
Together 7b	PI	32k	**7.64**	>100	>100	>100
+AdaGroPE	PI	32k	**7.64**	**7.58**	**7.58**	**7.60**

Performs consistently on more challenging tasks and integrates well with training-based methods

Analysis: Latency, Memory and Hyperparameter

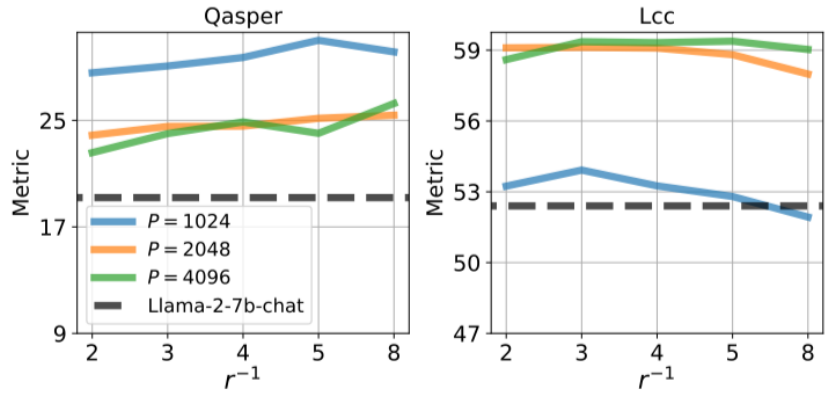
Latency and Memory Consumption

Model	Latency (s/token)			
	32k	64k	96k	128k
Llama-2-7b-chat	1.81	5.30	10.62	17.77
AdaGroPE-Llama-2-7b-chat	2.10	5.80	11.24	18.48

Model	Memory (MB)			
	32k	64k	96k	128k
Llama-2-7b-chat	28787	43925	59063	74193
AdaGroPE-Llama-2-7b-chat	29543	45433	61323	77201

≤10% latency & memory increase on average

Ablation Studies on the Hyperparameter Selection



Model	P	TOFEL (3k∼5k)	QuALITY (4k∼9k)	Coursera (5k∼17k)	SFiction (6k∼27k)
Llama-2-7b-chat	N/A	51.67	37.62	29.21	60.15
+AdaGroPE	1k	**61.34**	36.63	**37.50**	60.15
	2k	56.13	**38.12**	35.47	**64.06**
	4k	53.53	37.62	36.05	61.03

Default $r = 0.25$; smaller P works better for shorter contexts

Summary

- A novel positional encoding strategy, **AdaGroPE**, which extends the range of pre-trained position embedding by gradually increasing the reuse count based on the tokens' distance.

- A implementation of AdaGroPE in a dynamic, adaptive adjusted manner, maximizing the use of pre-trained position embedding in real-world scenarios with variable input lengths.

- Effective across various long-context benchmarks and LLMs. Experiment results show that AdaGroPE achieves state-of-the-art performance.

Thank You!

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